

Emerson León

Mathematician/ Software Developer

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EXPERIENCE

UNIVERSIDAD ANTONIO NARIÑO | MATH EDUCATION | AI OLYMPIADS

April 2024 – today | Bogotá, Colombia

- Researcher Profesor at Doctoral Studies of Math Education
- Organizer of the Colombian Olympiad in Artificial Intelligence and Team Leader of Colombia in international competitions IOAI
- Skills: teaching · math · artificial intelligence · python · pandas · sklearn · pytorch · html

EL CYBORG CHAMÁN | SOFTWARE DEVELOPMENT

June 2022 – July 2023 | Bogotá, Colombia

- Grant from Idartes for art, science and technology with Tercer Piso Danza
- Skills and tools: web development · arts and science · python · openAI · flask · django · JavaScript · artificial intelligence · three.js

LOS ANDES UNIVERSITY | POSTDOCTORAL RESEARCHER | MATH PROFESSOR

July 2015 – June 2017 | Bogotá, Colombia

- Research in discrete geometry, polytopes, Ehrhart theory, combinatorics
- Teaching: Linear algebra, differential, integral and multivariate calculus, discrete Geometry
- Skills and tools: discrete geometry · computational geometry · teaching · education · linux · web development · html · mathematical modeling · sagemath

COLOMBIAN MATHEMATICAL OLYMPIAD | ORGANIZING COMMITTEE | TEACHER

Jun 2002 – Dec 2006 | Bogotá, Colombia

- Coorganizer of the yearly math competitions for the whole country
- Helped to prepare talented Colombian children in math problem solving to participate in international competitions
- Skills and tools: math teaching · event organizing · problem solving · design of new math problems · creativity · algebra · number theory · geometry · combinatorics

EDUCATION

FREIE UNIVERSITÄT, BERLIN | DOCTOR RER. NAT. MATHEMATICS

Jan 2007 - Feb 2015 | advisor: Günter M. Ziegler

- Research in discrete geometry, mathematical modeling, topological spaces, moduli spaces, multidimensional geometry.
- In the thesis we defined and analyzed a mathematical object that parameterized all possible ways to divide the d-dimensional space into n convex regions.
- Scholarship Phase I and Phase II by Berlin Mathematical School.

UNIVERSIDAD NACIONAL DE COLOMBIA, BOGOTÁ | BACHELOR'S IN MATHEMATICS

Jul 2001 - Sep 2006

- Tuition reimbursement due to top grades each semester.
- Honors degree

SKILLS

PROGRAMMING

Proficient:

python · JavaScript · html · css · sagemath · L^AT_EX

Familiar:

C++ · matlab · Julia

LIBRARIES/Frameworks

flask · django · three.js · openAI · pandas · sklearn · matplotlib · pygame · sqlalchemy · node.js · tensorflow · sagemath

LANGUAGES

Spanish · English · German

PUBLICATIONS

- Binomial Inequalities of Chromatic, Flow, and Ehrhart Polynomials, with Matthias Beck *Disc. Comp. Geometry*, 2021
- Spaces of Convex n-Partitions *New trends in intuitive Geometry*, 2017
- Combinatorial Relationship Between Finite Fields and Fixed Points of Functions Going Up and Down *preprint*, 2023

AWARDS

SILVER+2 BRONZE MEDALS

INTERNATIONAL MATH OLYMPIAD IMO
Rumania 1999 | Korea 2000 | USA 2001

GOLD MEDAL

IBEROAMERICAN MATH OLYMPIAD
First place Cuba 1999

REFERENCES

Luis Serrano, AI scientist,
Serrano Academy

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Federico Ardila, Math
Professor, San Francisco State
University

 federico@sfsu.edu